

Basics Interview Question (Technical)

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Q1 Why C is a procedural language.

Ans:- Because C programs follow a procedure of steps written in it called functions. It follows a top to down approach i.e much importance is given to flow of program rather than on data on which function operates. On the other hand Java/C++ are object oriented language. They have bottom up approach.

Q2:- Why do you use `stdio.h` before program.

`stdio` stand for standard input output. `#include` is like including this standard input output library for your part of code. This way you can use the IO function available viz. `printf` & `scanf`. Since this is done before your code execution started. It's also called as preprocessor directive.

Q3:- Local variable & Global variable.

* Local Variable

A variable that is declared inside a function of a computer program is local variable.

Accessible only within the function it is declared.

Created when the function starts executing & it is destroyed when execution is complete.

More reliable & secure since the value cannot be changed by other function.

* Global Variable:-

A variable that is declared outside the function of a computer

Accessible by all the function in the program.

Remains in existence for the entire time the program is executing.

Accessible by multiple function therefore its value can be changed.

29) Call by Value Call by Reference

i) A copy of variable is passed. A variable itself is passed.

ii) Change in a copy of variable doesn't modify the original value of variable. Change in a copy of variable modify the original value of variable.

iii) Function name (variable-name1, variable-name2) Function name (&variable-name1, &variable-name2).

iv) Primitive type are passed using call by value. Objects are implicitly passed using call by reference.

Q5:- Recursion in C

Recursion is the process of repeating item in a self similar way. In Prog languages, if a program allows you to call inside the same function, then it is called a recursive call of the function.

```
void recursion() {  
    - recursion(); /* function calls itself */  
}  
  
int main() {  
    recursion();  
}
```


(Bits measures speed)
~~1 Bit~~ 1 Byte = 8 Bits
1 KB = 1024 Bytes

The C programming language supports recursion i.e. a function to call itself. But while using recursion, programmers need to be careful to define an exit condition from the function, otherwise it will go into an infinite loop.

Q6. What is pointer variable

The pointer in C language is a variable which stores the address of another variable. The variable can be of type int, char, array, function or any other pointers.

The size of the pointer depends on the architecture. However, in a 32 bit architecture, the size of pointer is 2 byte.

Consider the following

```
int n = 10;
```

```
int *p = &n; //
```

Pointer can be declared using *

```
int *a; // Point of int
```

```
char *c; // pointer of char
```


Advantages of pointer

- 1) Pointer reduces the code & improves performance, it is used for retrieving strings & trees etc & used with array structure & functions.

Q7) What is C array

An array is defined as the collection of similar type of data items stored at contiguous memory locations. Arrays are derived data types in C programming language which can store primitive type of data such as int, char, double, float etc.

It also has the capability to store collection of derived data types such as pointers, structures

C array is beneficial if you have to store similar element.

If you want to store marks of a student in 6 subject. then we don't need to define different variable for the marks in different subject.

Instead we can define an array which can store the marks in each subject at contiguous memory location.

Q8 What is DataBase & Database management system.

Data Base :- It is the collection of inter related data which to retrieve, insert and delete data efficiently.

It is also used to organise the data in the form of a table, schema, views & report.

Data Base Management System.

It is a software which is used to manage data bases e.g. MySQL, Oracle etc are very popular database which is used in different applications.

DBMS provides an interface to perform various operations like database creation, storing data in it, updating data, creating a table in database & a lot more.

It provides protection and security to the data bases. In case of multiple users it also maintains data consistency.

DBMS provide follows users the following tasks.

- Data Definition :- It is used for creation, modification & deletion of actual data in the database.
- ② Data Updation :- It is used for insertion, modification & deletion of actual data in the databases.
- ③ Data Retrieval :- It is used to retrieve the data from the database which can be used by applications for various purposes.
- ④ User Administration :- It is used for registering & monitoring users, maintain data integrity, dealing with concurrency control, monitoring performance & recovering information corrupted by unexpected failure.
- ① Data Definition :- It is used for creation, modification, and removal of definition that defines the organisation of data in the databases.

Characteristics of DBMS

- DBMS contains automatic backup & recovery procedures.
- It can reduce the complex relationship between data.
- It is used to provide security of data.
- It is used to support manipulation & processing of data.

Advantages:-

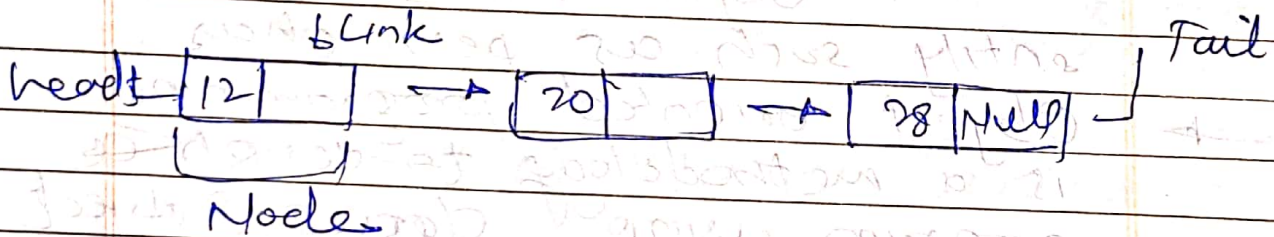
- Control Data Redundancy (Repetition Control)
- Data Sharing / Improved Data Access
- Easily Maintenance
- Reduce Time
- Backup
- Multiple User Interface
- Improve Data Security / Data Access Decision Making & end user productivity

Q9 Explain Functionality of linked list

A linked list consists of two parts
Information part & link part

It is the collection of objects called nodes that are randomly stored in memory

A node consists of two fields. i.e
Data stored at that particular address
& the pointer which contains the
address of the next node in the
memory



Q10 Normalisation

Normal is the process of organizing
the data in the databases.

Normalization is used to minimize
the redundancy from a relation or set
of relations

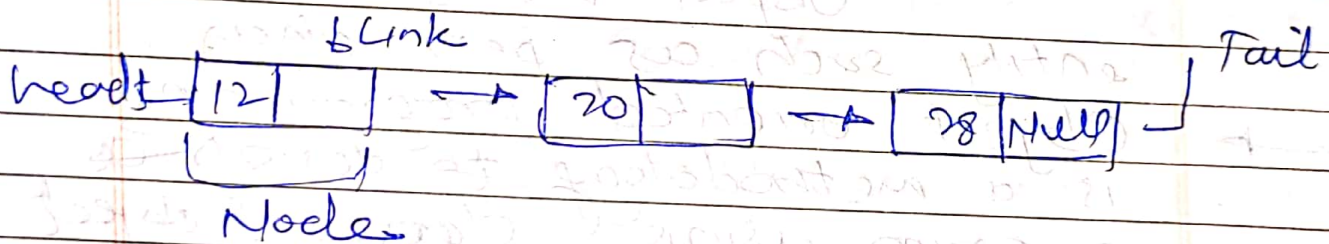
Normalization divides the larger table
into smaller table & links them using
relationship.

Q9 Explain Functionality of linked list

A linked list consists of two parts: Information part & link part.

It is the collection of objects called nodes that are randomly stored in memory.

A node consists of two fields: i.e. Data stored at that particular address & the pointer which contains the address of the next node in the memory.



Q10) Normalisation

Normal is the process of organizing the data in the databases.

Normalization is used to minimize the redundancy from a relation or set of relations.

Normalization divides the larger table into smaller table & links them using relationships.

1NF, 2NF, 3NF & 4NF

Q# C++ Object oriented programming

The major purpose of C++ programming is to introduce the concept of Object orientation to the C progr.

Object oriented programming is a paradigm / pattern that provides many concept such as inheritance, data binding & polymorphism.

OOPS :-

Object means a real world entity such as pen, chair.

→ Object oriented programming is a methodology to design a program using classes & object.

Object :- Any entity that has state & behaviour is known as object. It can be physical & logical.

Class :- Collection of objects is called class. It is a logical entity.

Inheritance :- When one object acquires all the properties & behaviours of parent object. known as inheritance. It provides code reusability.

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Polymorphism :- When object acquires all the properties & behaviours of parent object.

Polymorphism :- When one ~~object~~ task is by different ways i.e known as polymorphism. For e.g

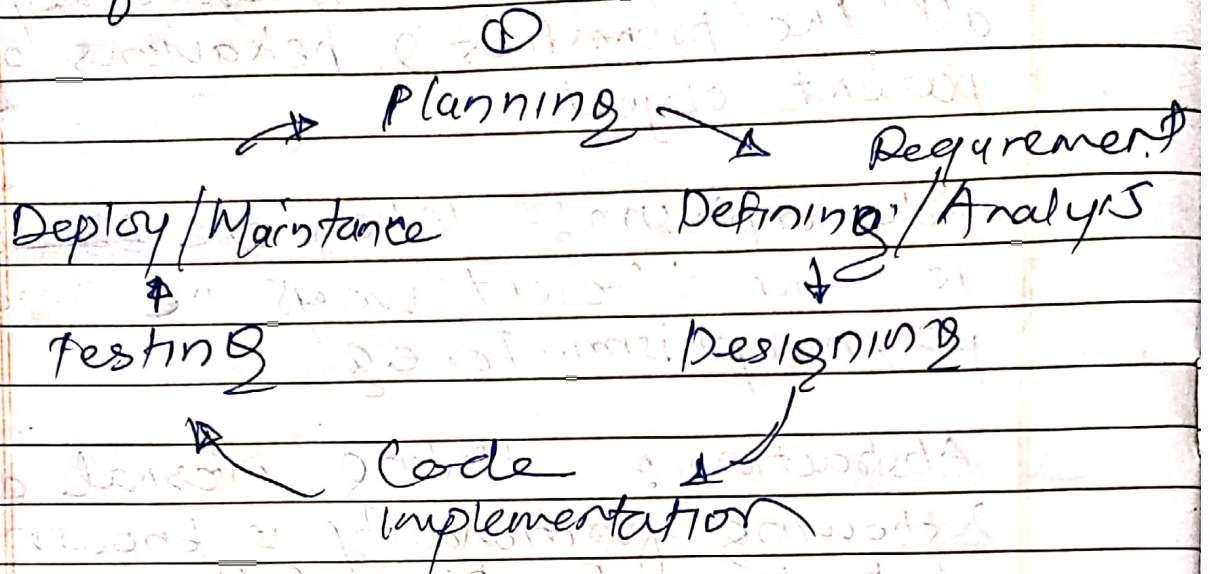
Abstraction :- Hiding internal details & showing functionality is known as abstraction. E.g Phone Call, we don't know the internal processing.

Encapsulation :- Binding or (wrapping) code & data together into a single unit is known as encapsulation.

Advantages of OOPS over Procedure Oriented

- 1) OOPS makes development & maintenance easier where as in procedure-OPS it is not easy to manage if code grows as project size grows.
- 2) OOPS provide data hiding whereas POPs a global data can be accessed from anywhere
- 3) OOPS provide ability to simulate real world event much more effectively. We can provide the solution of real world problem if we are using OOPS

Q2 Software Development Life Cycle



A Life cycle model represents all the methods required to make a software product transit through its life cycle stages.

Stage 1 :- Planning :-

- Requirement Analysis
- Planning of

a) Quality Assurance Requirement

b) Identification of Risk associated.

- Software Requirement Specification Document which contain all the project requirement to be constructed & developed during the project life cycle

Stage 2 :- Defining Requirement

- Document acceptance from Stakeholder
- It is done by SRS

Stage 3:- Designing the Software

- Bring down all the knowledge of requirement analysis, and design of the software project.

Stage 4:- Developing.

In SDLC, Actual Development begins
 & programming is built

Stage 5:- Testing

After the code is generated, it is tested against the requirements to make sure the products are solving the needs addressed & gathered during requirement stage.

Stage 6:- Deployment

Once the software is certified, & no bugs & error are stated. then it is deployed.

Stage 7:- Maintenance

One when the client starts using the developed systems, then the real issues come up & requirement to be solved from time to time.

213 Embedded C

① Used to create applicatⁿ → C is used for writing which is based on application program & system microprocessor & program processor e.g. Display Bo^{ard} mathematical or Scientific

② Hardware knowledge is required. Hardware knowledge not required

③ Memory management should be efficient. No need to take care about memory using efficiently.

④ Hardware Dependent Hardware independent

⑤ Compiler generates hardware dependent file. Compiler generates OS dependent file

⑥

Transmission Control

Uses Datagram

- 1) It is a connection oriented protocol, which means, connection need to be established before data transfer. It is a connectionless protocol which it sends the data without checking.
- 2) TCP is reliable & provide assurance for delivery of the data packet. It is unreliable as it does not guarantee the delivery of data packets.
- 3) TCP is slower. UDP is faster.
- 4) TCP size is 20 Bytes. UDP size is 8 Bytes.
- 5) It is flow control mechanism in which too many packets can't be sent to receiver at a same time. Does not follow such mechanism.
- 6) Perform error checking by using checksum. Does not perform any error checking.
- 7) Used where secure & reliable communication is required. like military services, web browsing & email. Uses where fast communication required. & does not care about reliability like game streaming & video & music streaming.

Q14 C Strings

The string can be defined as the one dimensional array of characters terminated by a null '\0'. The character array or the string is used to manipulate text as word or sentence. Each character in the array contains one byte of memory. & the last character must be zero.

Q15: Class

Structure

- | | |
|--|---|
| ① Class is collection of data member & member function | ① Structure is a collection of data variable (members). |
| ② High security | ② Less security |
| ③ Class members are private by default | ③ Structure members are public by default |
| ④ Class is used in large program | ④ Structure is used in small program |
| ⑤ Class | ⑤ Struct |
| ⑥ Class can be inherited | ⑥ Structure can't be inherited |
| ⑦ Class occupies more space | ⑦ Structure occupies less space. |

C language to swap two numbers
with using thread.

```
#include <stdio.h>
```

```
#include <conio.h>
```

```
void main()
```

```
{
```

```
int i, j;
```

```
printf("Enter the value of i:\n");
```

```
scanf("%d", &i);
```

```
printf("Enter the value of j:\n");
```

```
scanf("%d", &j);
```

```
printf("value of i before swap: %d\n", i);
```

```
printf("value of j before swap: %d\n", j);
```

```
i = i + j;
```

```
j = i - j;
```

```
i = i - j;
```

```
printf("value of i after swap: %d\n", i);
```

```
printf("value of j after swap: %d\n", j);
```

```
}
```


DBMS

RDBMS

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- ① Stores Data in the form of a file
 - ② Hierarchical arrangement of data
 - ③ Allow one user at a time
 - ④ Manages data in a computer
 - ⑤ Cannot be normalized
 - ⑥ Can't handle large amount of data
- ① Stores data in the form of table
 - ② Stores data in the form of rows & column within table
 - ③ Allow more than one user at a time
 - ④ Maintains the relationship of tables in a database
 - ⑤ Can be normalized
 - ⑥ Able to handle large amount of data

⑦

Relational.

Q Python String

Collection of character surrounded by single, double or triple quotes

```
str = "fawwad"
```

```
str[0] = '012345'
```

② Python List

List is used to store sequence of various types of data types. Python list is mutable type that means we can modify its element after it is created.

Items separated with commas & enclosed with the square bracket `[]`

```
L1 = ["fawwad", 102, "USA"]
```

```
L2 = [1, 2, 3, 4, 5, 6]
```

③ Python Tuple

Python Tuple is used to store the sequence of immutable Python objects. Tuple is immutable, value of items stored in tuple can't be changed.

Tuple can be written as collection of comma-separated (,) values enclosed with () brackets

```
Tuple1 = (1, 2, 3, 4, 5)
```


Python Set :-

A python set is the collection of unordered set. Each element in the set must be unique, immutable & the set removes the duplicate elements. Sets are mutable, it means we can modify it after its creation.

A set can be created by enclosing the comma separated immutable items with curly braces.

```
Days = { "Mon", "Tue", "Wed", "Thu", }
```

Python Dictionary :-

Python dictionary is used to store the data in a key value pair format.

Dictionary can be created by using multiple Key value pairs enclosed with the curly bracket {}, & each key is separated from the its value by the (:) colon.

```
Dict = { "Name": "Famrad", "Age": 22 }
```


LAN Local Area Network

Local Area Network :-

- Local area network is a group of computers connected with each other in a small places such as school, hospital, apartment etc.
- LAN is secure because there is no outside connection.
- LAN due to their small size are considerably faster, their speed can range anywhere from 10 to 100 MBPS.
- LANs are limited to wire connection, there is a new evolution to the LANs that allows LAN on a wireless connection.

MAN (Metropolitan Area Network)

- MAN network covers larger area by connections LANs to a larger network of computers (small towns, cities).
- In MAN various local area networks (LANs) are connected to each other through telephone lines.
- The size of MAN is larger than LANs & smaller than WANs.

* WAN (Wide Area Network)

- Wide area network provides long distance transmission of data
- WAN size is larger than LAN & MAN
- Internet Connection is an example of WAN
- Other example of WAN are Broadband connections such as 3G, 4G, etc.

* OSI (Open System Interconnections)
(Pratik do not teach sasta presentation Application)
A conceptual framework that describes the function of a networking or telecommunication system.

- ① Physical layer: (Electrical & physical)
- ② Data link layer: Node to Node Transfer
- ③ Network layer: Router functionality
- ④ Transport layer: Data transfer System to host
- ⑤ Session layer: -
- ⑥ Presentation layer: Syntax that application accept
- ⑦ Application layer: web browser (Google Chrome)

Q

* Phases 15 C

Source
file (.c or .cpp...)

Preprocessing

Pre-processed file as input (.i) #codes

Compilation

High level source code is converted into assembly language. (.s)

Assembly Code

Assembly

Converts assembly instruction resulted from compilation & generates object code

(.o or .obj)

Machine code

1

Linking

Takes static libraries & object file & generates single executable.

.out, .exe
executable